Homework-03

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Github repository: https://github.com/leahlamarre-ship-it/ENVS-193DS_homework-03

```
library(tidyverse) #loading in packages
library(janitor) #loading in janitor
library(here) #loading in here
library(readxl) #loading in readxl

salinity <- read_csv("/Users/leahpettway/ENVS-193DS/github/ENVS-
193DS_homework-03/HW3_data/salinity-pickleweed (1).csv") #loading in the data

my_data <- read_csv("/Users/leahpettway/ENVS-193DS/github/ENVS-
193DS_homework-03/HW3_data/Personal_data.csv") #loading in personal data

my_data_clean <- my_data |> # Creating the data frame to clean up data
  clean_names() #cleaning the whole my_data file
```

Part 2. Problems

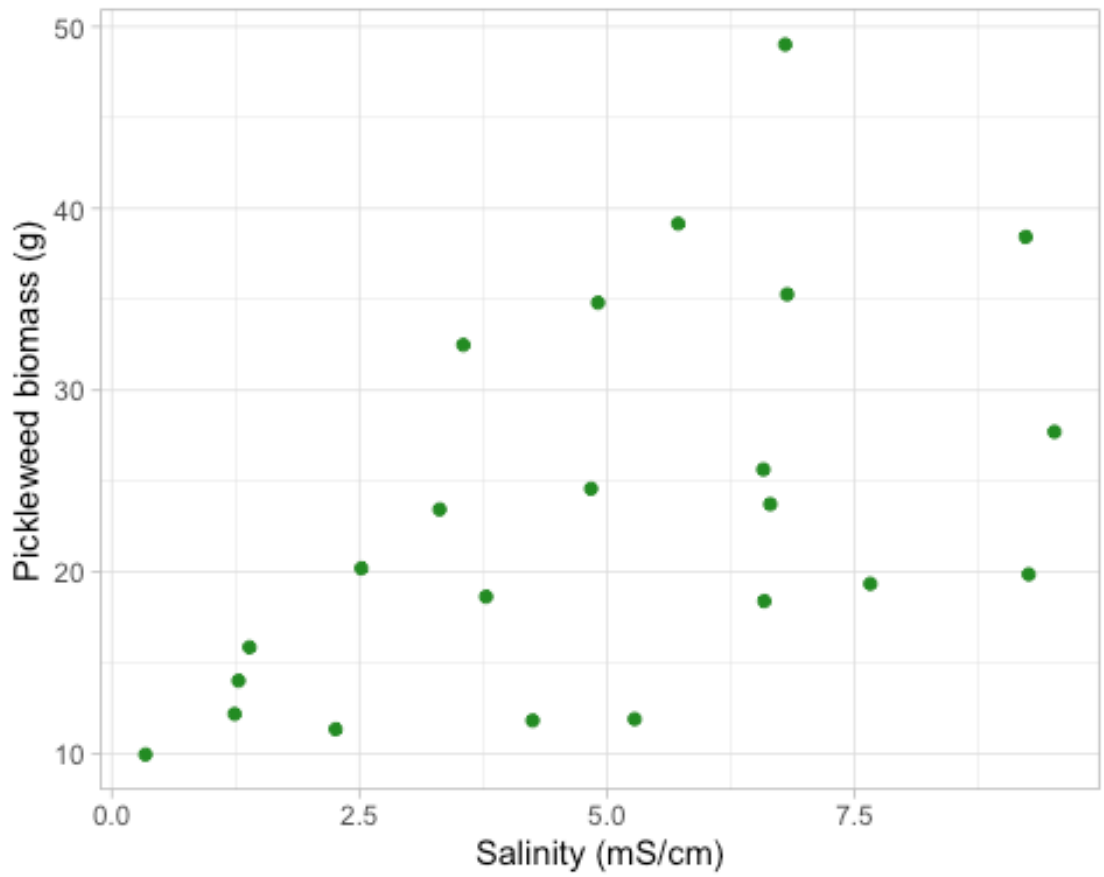
Problem 1. Slough soil salinity

a. An appropriate test

The appropriate tests to determine the strength of the relationship between salinity and California pickleweed biomnass is a pearson and spearman correlation test since both tests how two variables are related to eachother or their strength of correlation. Pearson does this with variables that are continuous while spearman follows nominal or ranked data .

b. Create a visualization

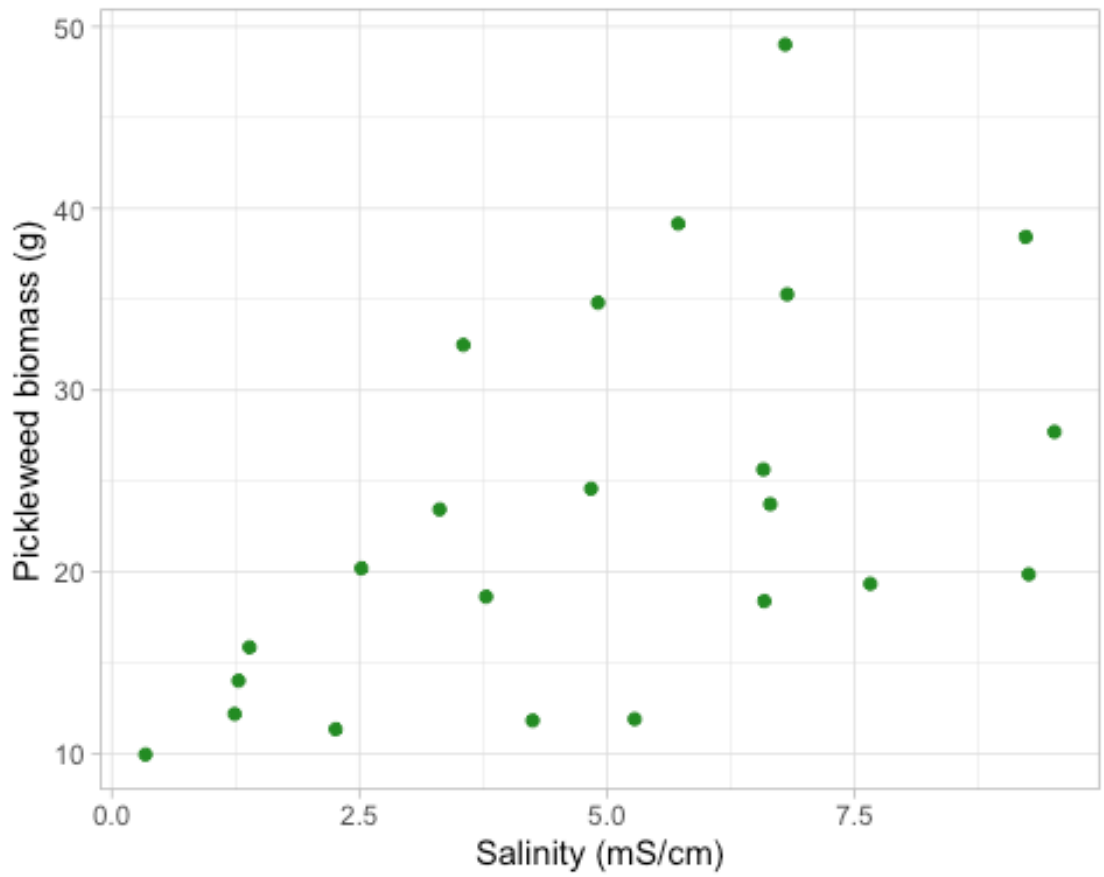
```
salinity_clean <- salinity |> #new object created from salinity data frame
  clean_names() |> # cleaning up names so all are lowercase and no spaces
  rename("pickleweed_g" = "pickleweed") # including the unit pickle weed is
in

ggplot(data = salinity_clean, #use salinity_clean data frame
       aes(x = salinity_m_s_cm, #make the x-axis salinity_m_s_cm
           y = pickleweed_g)) + # make the y-axis pickleweed_g
  geom_point(color = "forestgreen") + # add point geometry
  labs(x = "Salinity (mS/cm)", #relabel the x-axis
       y = "Pickleweed biomass (g)") + # relabel the y-axis
  theme_light() #change theme from default
```



C. Checking Assumptions

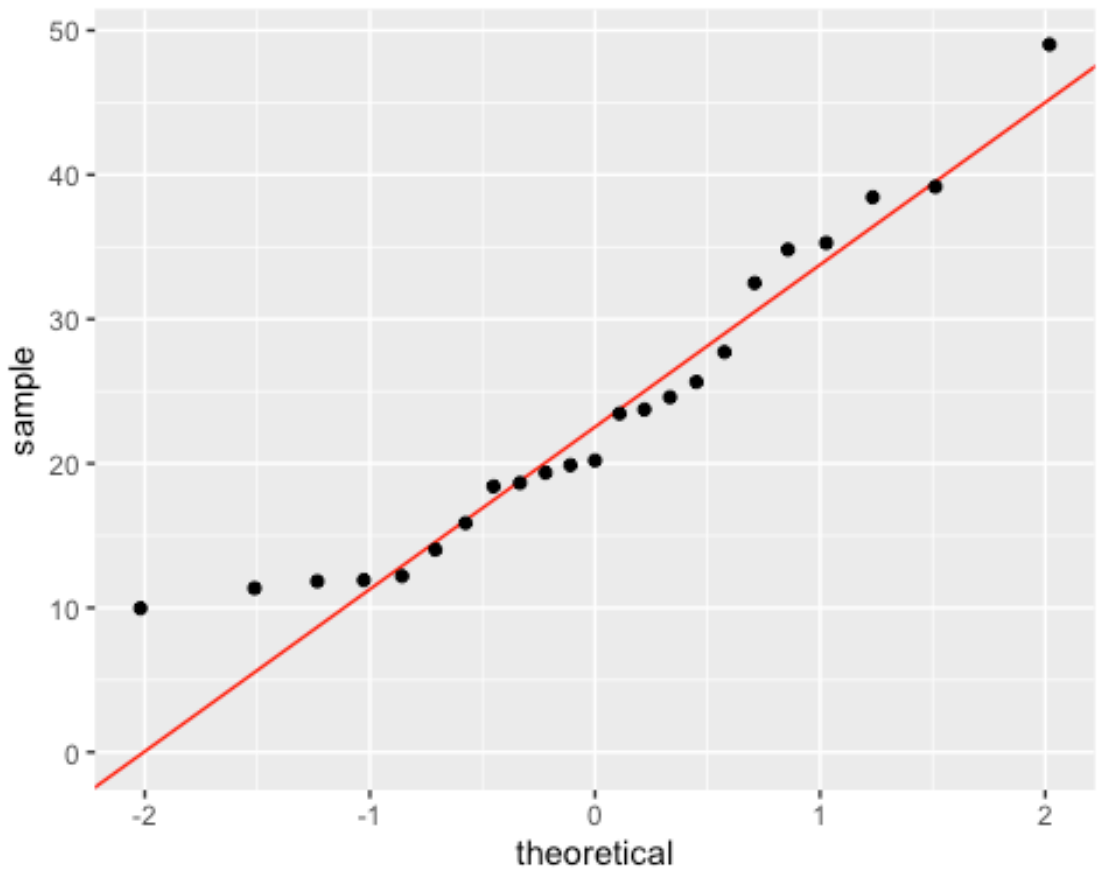
Assumption of a linear relationship

```
ggplot(data = salinity_clean, #use salinity_clean data frame
       aes(x = salinity_m_s_cm, #make the x-axis the predictor variable
           y = pickleweed_g)) + #make the y axis the response variable
  geom_point(color = "forestgreen") + #add point geometry to plot colored
different from the default
  labs(x = "Salinity (mS/cm)", #relabel the x-axis
       y = "Pickleweed biomass (g)") + #relabel the y- axis
  theme_light() # change theme from default
```

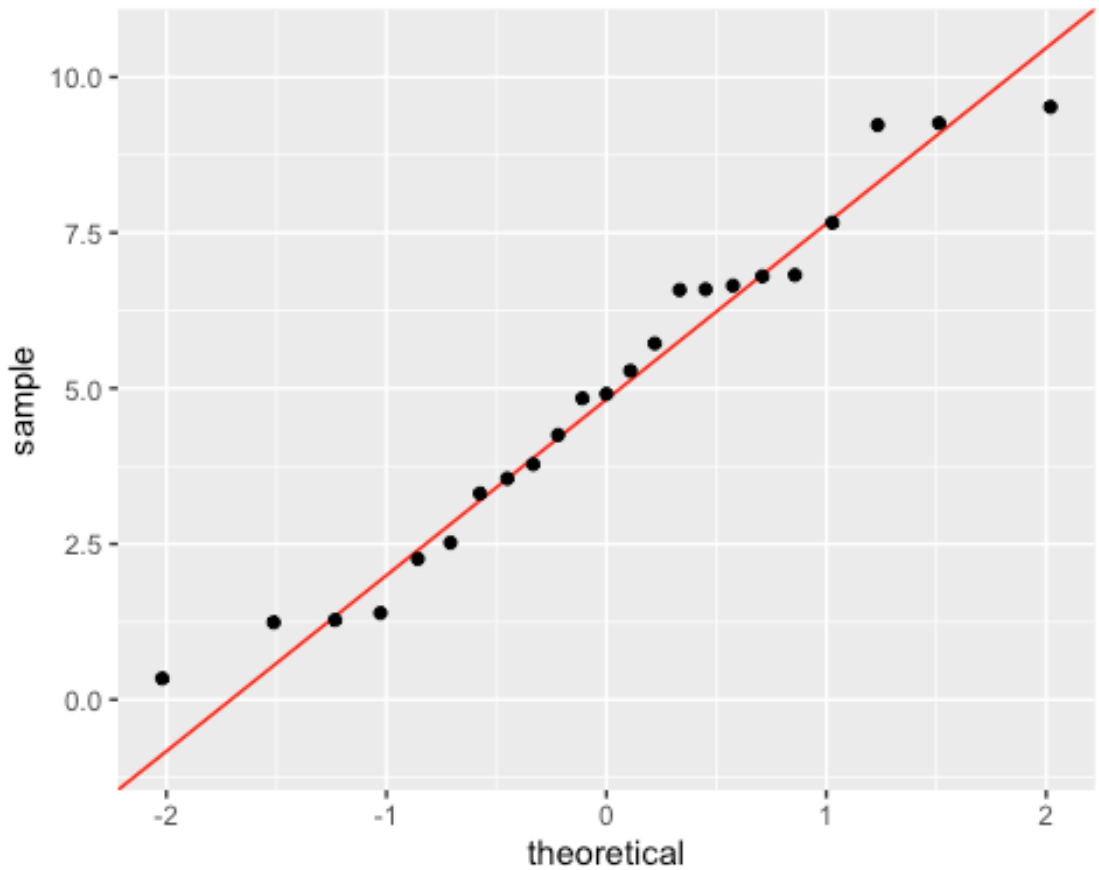


Assumption of normal distribution

```
# base layer: ggplot call
ggplot(data = salinity_clean, # starting data frame using salinity_clean
       aes(sample = pickleweed_g)) + # y-axis for QQ plot
  # first layer QQ reference line
  geom_qq_line(color = "red") + # using red so line is easier to see
  # second layer: QQ plot
  geom_qq()
```




```
# base layer: ggplot call
ggplot(data = salinity_clean, # starting data frame using salinity_clean
       aes(sample = salinity_m_s_cm)) + # y-axis for QQ plot
  # first layer QQ reference line
  geom_qq_line(color = "red") + # using red so line is easier to see
  # second layer: QQ plot
  geom_qq()
```



The assumptions I needed to check a few assumptions for a pearson correlation test, which is a linear relationship between the variables and a normal distribution of the variances for each variable. For the linear relationship I visually checked the data once I formatted it into a scatterplot to see the ralitonship between the two variables which appears to be linear due to the general trend of the points. In checking for normal distribution I created two qqplots to see the distribution of the variences and in both plots the points and reference line were quite aligned.

```
cor.test(salinity_clean$salinity_m_s_cm, salinity_clean$pickleweed_g,
         method = "pearson") # using the salinity_clean for a dataframe and
then choosing both columns for the pearson's R correlation test


    Pearson's product-moment correlation

data:  salinity_clean$salinity_m_s_cm and salinity_clean$pickleweed_g
t = 2.8979, df = 21, p-value = 0.008605
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1568265 0.7757682
sample estimates:
      cor
0.5344778
```

d. Results communication

To evaluate the strength of soil salinity and pickle weed biomass I used a Pearson's correlation test, which I came to the conclusion to use since both variables are continuous and after subsequent checks the data seems to have a linear relationship and normal distribution of variance. The result of the Pearson's R test is that we found a moderate relationship between soil salinity and pickle weed biomass(Pearson's $r = 0.53$, $t(21) = 2.9$, $p = 0.01$, $\alpha = 0.05$)

e. Test implications

In the results of the correlation test, a moderate relationship was found between the variables soil salinity and pickle weed biomass. These results suggest that while soil salinity may have some affect on the pickle weed biomass it may not be the ultimate determinant of the biomass. That means for the best success in pickle weed some soil salinity is good but making sure no extremes in soil salinity are emphasized in planting pickle weed.

f. Double check your own work

```
cor.test(salinity_clean$salinity_m_s_cm, salinity_clean$pickleweed_g,
         method = "spearman") # using the salinity_clean for a dataframe and
then choosing both columns for the Spearmancorrelation test


    Spearman's rank correlation rho

data:  salinity_clean$salinity_m_s_cm and salinity_clean$pickleweed_g
S = 824, p-value = 0.003426
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.5928854
```

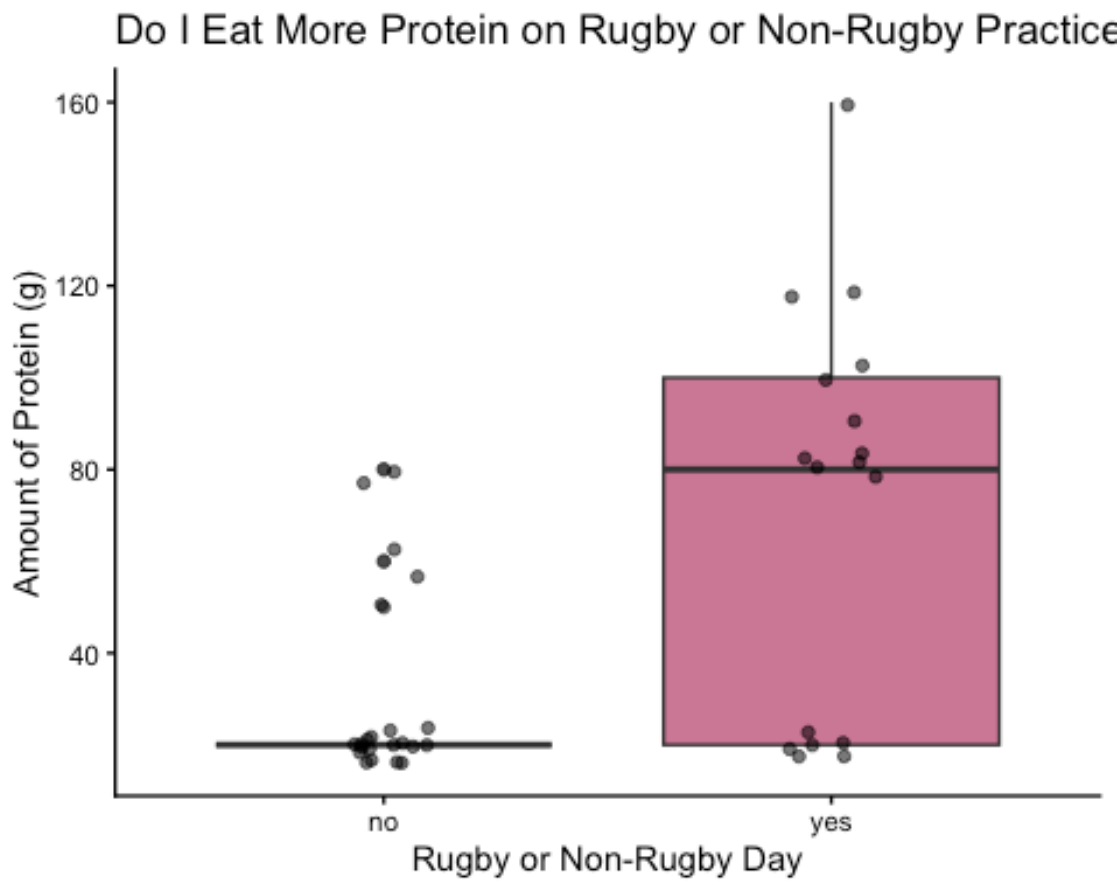
The two tests would've led me to the same decision that we would fail to reject the null hypothesis as well as finding a moderate relationship between the variables soil salinity and pickle weed biomass We did ultimately find a moderate relationship between soil salinity and pickle weed biomass (Spearman $\rho = 0.59$, $S = 824$, $p = 0.003$, $\alpha = 0.05$).

Problem 2. Personal data

a. Updating your visualizations

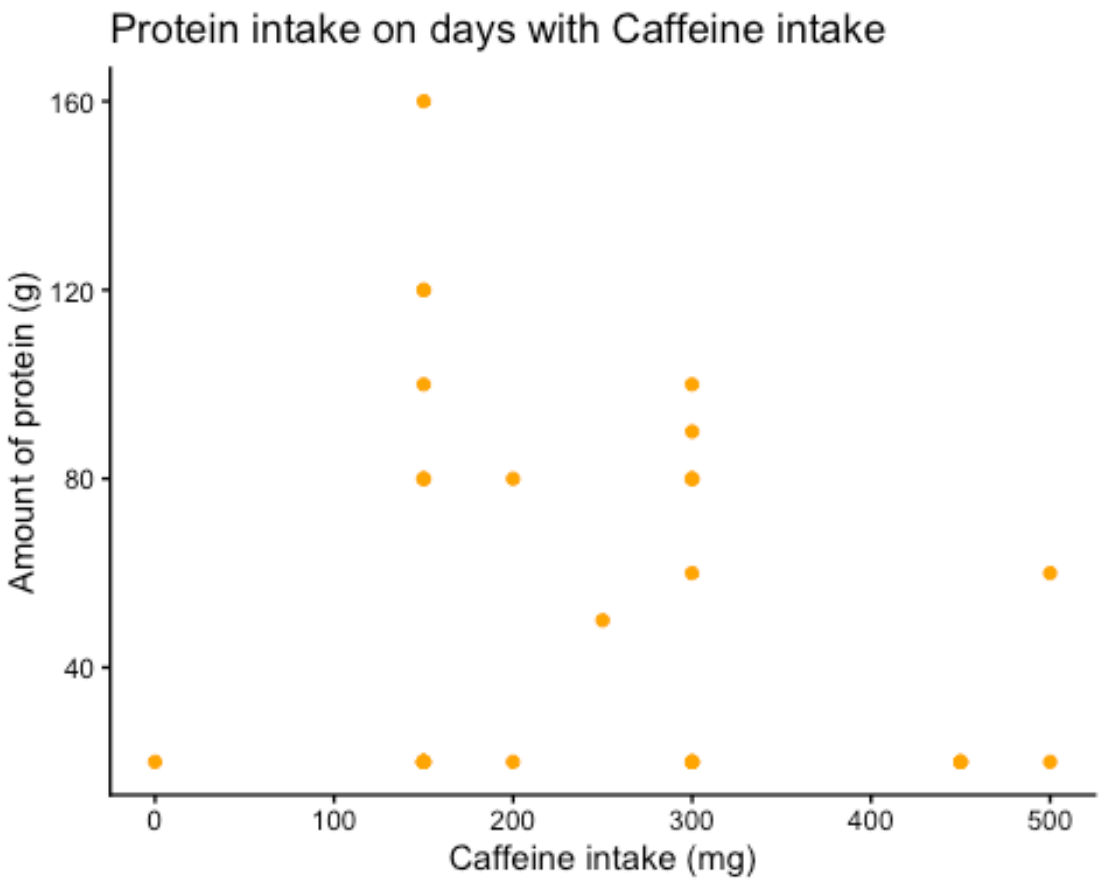
```
ggplot(data = my_data_clean, # using my data as the data frame
       aes(x = rugby_practice_day_yes_or_no, # setting the x axis as the type
of day
           y = amount_of_protein_grams, # setting the y axis as the amount of
protein
           fill = rugby_practice_day_yes_or_no)) + # adding color to the
boxplot
  geom_boxplot(alpha = 0.7) + # boxplot a little transparent to still see the
```

```
observations
  geom_jitter(width = 0.1, alpha = 0.6) + # jittering the points to stay
closer together
  labs(x = "Rugby or Non-Rugby Day", # relabeling the axis to be more clear
       y = "Amount of Protein (g)",
       title = "Do I Eat More Protein on Rugby or Non-Rugby Practice Days?")
+ # adding a title to the graph
  scale_fill_manual(values = c("yes" = "maroon",
                               "no" = "lightblue")) +
  theme_classic() +
  theme(legend.position = "none")
```




```
ggplot(data = my_data_clean, # using the cleaned my data for data frame
       aes(x = caffeine_intake_milligrams, # x-axis
           y = amount_of_protein_grams)) + # y-axis
  # first layer: showing the underlying data
  geom_jitter(height = 0, # no jitter in the vertical direction
              width = 0.1,
              color = "orange") +
  labs(x = "Caffeine intake (mg)", # labeling the axes for caffeine intake
and amount of protein
       y = "Amount of protein (g)",
       title = "Protein intake on days with Caffeine intake") +
```

```r
theme_classic() + # using a theme
theme(legend.position = "none") # getting rid of the legend
```



b. Captions

Figure 1. Protein intake varies between rugby practice days and non-practice days. Points represent observations of daily protein intake (g) recorded by the author on rugby practice days and non-practice days (n = 39). Each point represents one day of recorded intake. The plot compares the distribution of protein intake across the two categories to examine whether protein consumption differs depending on whether I participated in rugby practice that day.

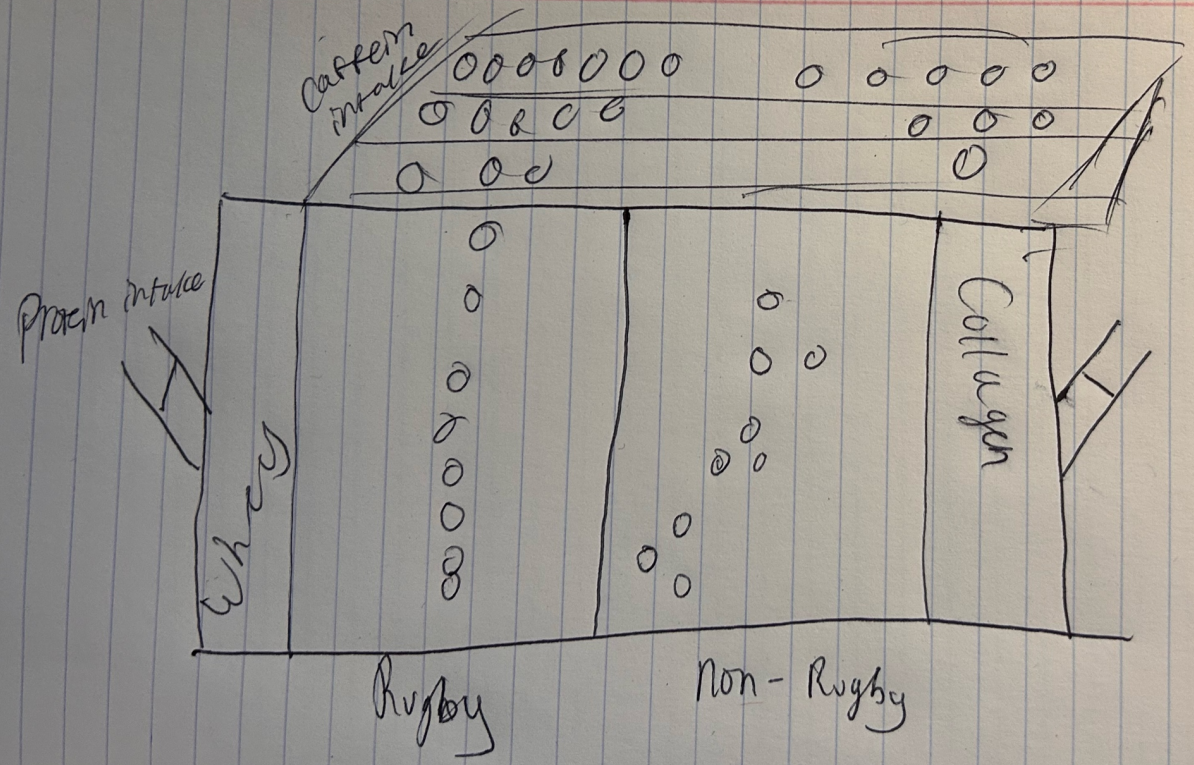
Figure 2. Protein intake tends to increase on days with higher caffeine consumption. Points represent observations of daily caffeine intake (mg) and daily protein intake (g) recorded by me (n = 39). Each point represents one day of personal dietary intake.

Problem 3.

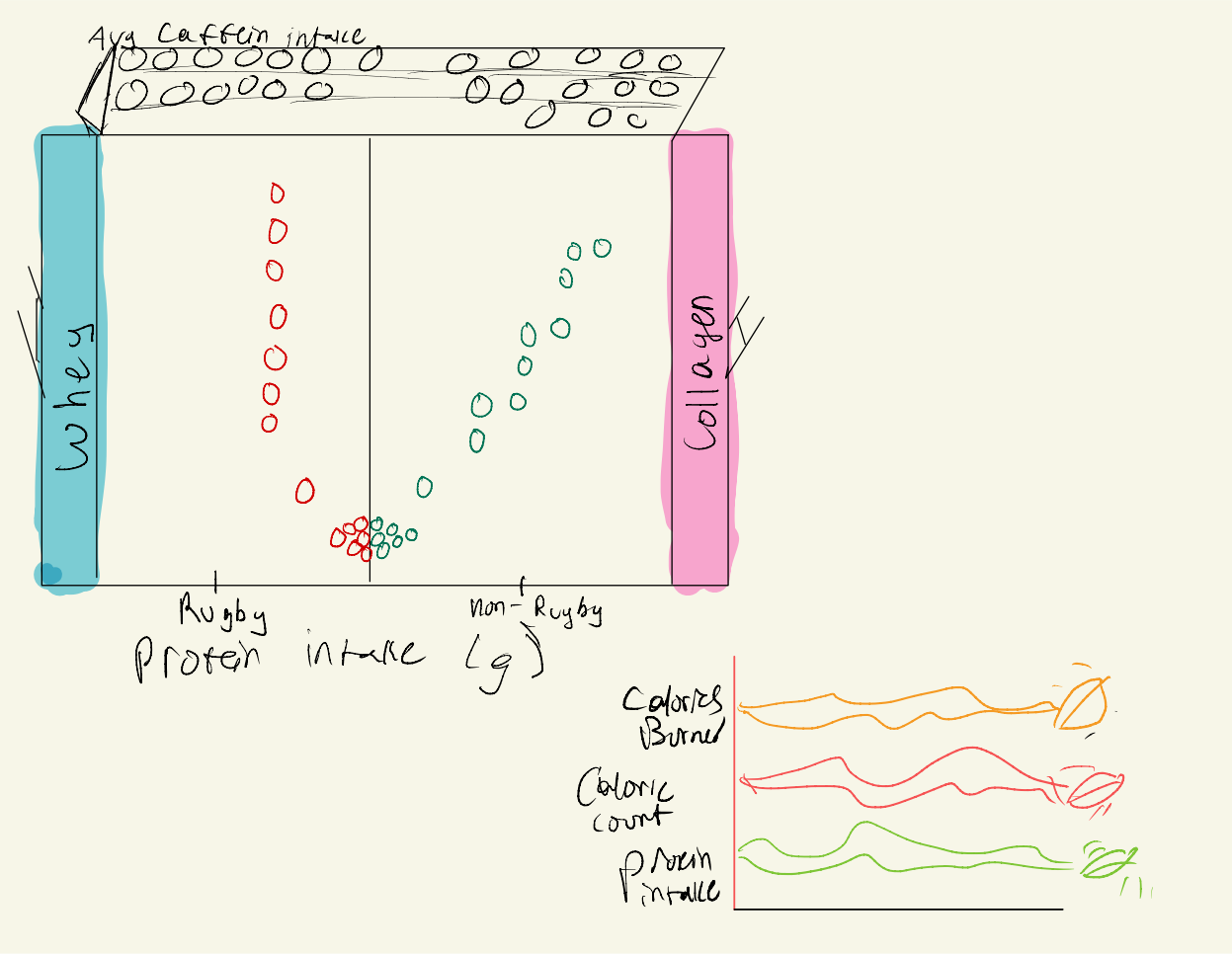
a. Describing what an affective visualization would look

An affective visualization for my personal data could potentially look like data formatted like a rugby pitch or creatively showcasing rugby and the protein intake intermixed. The usage of heatmaps is used in rugby strategy so potentially something similiar with my data formatted to resemble rugby in a way can hook a reader and pull them into my visualization. Further incorporating visuals of food since my data relates to protein and food often can catch a readers eyes if presented well.

b. Create a sketch of idea


Casein intake
Protein intake
Whey
Collagen
Rugby
Non-Rugby

c. Make a draft of your visualization



d. Write an artist's statement

The content of my piece is showing a rugby pitch but seperated ot represent the days rugby and non-rugby as two different teams as well as the points in defense and attacking shapes. I included tryzones which are the score areas as a type of protein from my data and I tried to make the stands my caffeine intake as the fans above the pitch(field). I found rugby the sport itself influential and took the look of the field as inspiration. The form of my work wsa a digital piece created on my ipad with a stylus.

e. Prep materials

https://docs.google.com/presentation/d/1bT6IQwot0-sYsC-0H2kMh1y3w-CA-Mhz9qoSS9oiYV4/edit?usp=sharing

Problem 4. Statistical critique

a. Revisit and summarize

The statistical test included is a linear-mixed model effects. The occurrence and abundance are the response variables, while year of sampling or year of construction was the predictor variable along with types of substrates, site, soil squares, and texture of soil.

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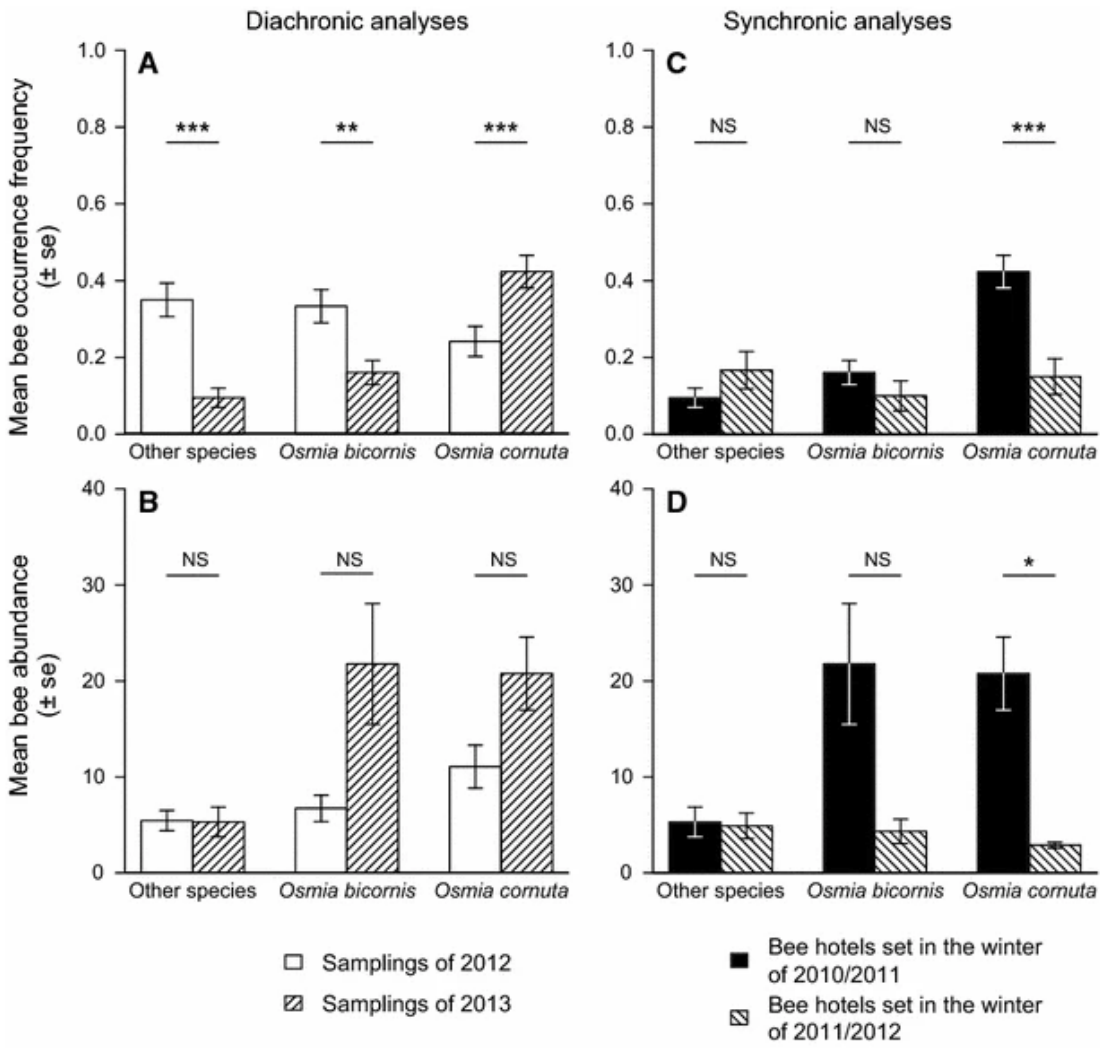

b.

The authors visually show the mean bee abundance and occurrence for multiple sampling sites and times while also including a standard error bar. The x and y axis are in logical places especially for containing multiple plots within one figure,along with the arrangement of the multiple plots. They ultimately do not show the underlying data.

c. Aesthetic clarity

The authors handled visual clutter decently well it still can be confusing at first to look at the graphs since there is multiple showing slightly different things. The data to ink ratio is pretty good with keeping the text to a minimum and mainly highlighting the actual data.

d. Recomendations

I would change the way the species names are represented as well as maybe changing the patterns used to differentiate the different years and species. I belive maybe shortening the names would help reduce text and confusion since a lot of the bee species names are very similiar, as well as finding patterns that vary mnore so the differnt columns dont look as similiar.